

ابزار دقیق

ویژگی های حسگرها و طبقه بندی حسگرها

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✓ ویژگی های استاتیکی

- صحت
- دقت
- تکرار پذیری
- بازتولید
- پایداری
- خطا
- نویز
- رانش
- وضوح
-

✓ ویژگی های دینامیکی (پویایی)

- سیستم درجه صفر
- سیستم درجه یک
- سیستم درجه دو
- پاسخ دهی و بازیابی

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2.3 Dynamic Characteristics

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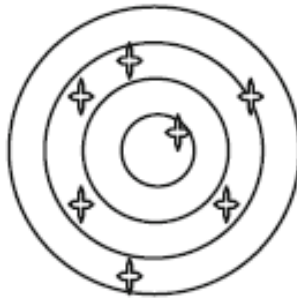


درستی (صحت)

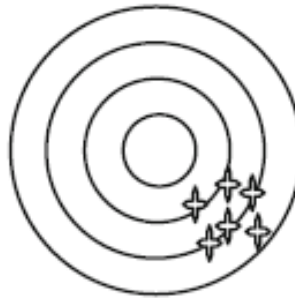
2.2.1 Accuracy

Accuracy of a sensing system represents the correctness of its output in comparison to the actual value of a measurand. To assess the accuracy, either the system is benchmarked against a standard measurand or the output is compared with a measurement system with a superior accuracy.

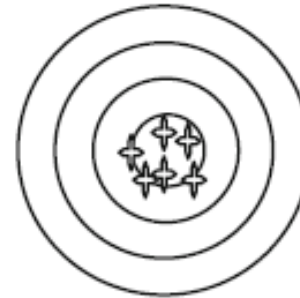
For instance considering a temperature sensing system, when the real temperature is 20.0°C , the system is more accurate, if it shows 20.1°C rather than 21.0°C .



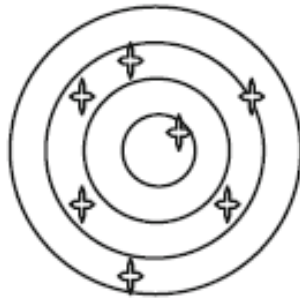
Low precision -
low accuracy



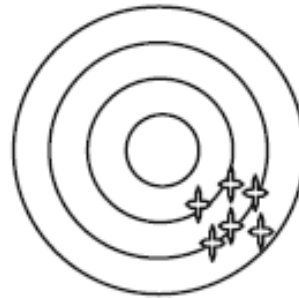
High precision -
low accuracy



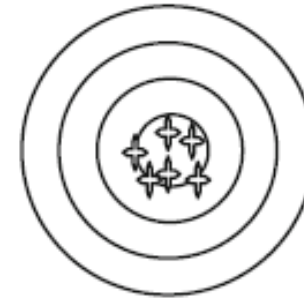
High precision -
high accuracy



Low precision -
low accuracy



High precision -
low accuracy



High precision -
high accuracy

2.2.2 Precision

دقت

Precision represents capacity of a sensing system to give the same reading when repetitively measuring the same measurand under the same conditions. The precision is a statistical parameter and can be assessed by the standard deviation (or variance) of a set of readings of the system for similar inputs.

2.2.3 *Repeatability*

When all operating and environmental conditions remain constant, *repeatability* is the sensing system's ability to produce the same response for successive measurements. Repeatability is closely related to precision. Both long-term and short-term repeatability estimates can be important for a sensing system.

For a temperature sensing system, when ambient temperature remains constant at 21.0 °C, if the system shows 21.0, 21.1, and 21.0 °C in 1 min intervals, and shows 22.0, 22.1, and 22.2 °C after 1 h, in similar 1 min intervals, the system has a good short-term and poor long-term repeatability.

2.2.4 *Reproducibility*

Reproducibility is the sensing system's ability to produce the same responses after measurement conditions have been altered.

For example, if a temperature sensing system shows similar responses; over a long time period, or when readings are performed by different operators, or at different laboratories, the system is reproducible.

2.2.5 *Stability*

Stability is a sensing system's ability to produce the same output value when measuring the same measurand over a period of time.

2.2.6 Error

Error is the difference between the actual value of the measurand and the value produced by the sensing system. Error can be caused by a variety of internal and external sources and is closely related to accuracy. Accuracy can be related to *absolute* or *relative error* as:

$$\begin{aligned} \text{Absolute error} &= \text{Output} - \text{True value}, \\ \text{Relative error} &= \frac{\text{Output} - \text{True value}}{\text{True value}}. \end{aligned} \quad (2.1)$$

2.2.7 Noise

The unwanted fluctuations in the output signal of the sensing system, when the measurand is not changing, are referred to as *noise*. The standard deviation value of the noise strength is an important factor in measurements. The mean value of the signal divided by this value gives a good benchmark, as how readily the information can be extracted. As a result, signal-to-noise ratio (S/N) is a commonly used figure in sensing applications. It is defined as:

$$\frac{S}{N} = \frac{\text{Mean value of signal}}{\text{Standard deviation of noise}} \quad (2.2)$$

Noise can be caused by either *internal* or *external* sources. Electromagnetic signals such as those produced by transmission/reception circuits and power supplies, mechanical vibrations, and ambient temperature changes are all examples of external noise, which can cause systematic error. However, the nature of internal noises is quite different and can be categorized as follows:

The magnitude of a thermal noise in a resistance of magnitude R (Ω) is extracted from thermodynamic calculations and is equal to:

$$\bar{v}_{\text{rms}} = \sqrt{4kTR\Delta f}, \quad (2.3)$$

in which $\bar{v}_{\text{rms}}$ is the root-mean-square of noise voltage, which is generated by the frequency component with the bandwidth of Δf , k is the Boltzmann's constant, which is equal to $1.38 \times 10^{-23} \text{ JK}^{-1}$, and T is the temperature in Kelvin.

Example 1. The rise and fall time of a sensor signal are generally inversely proportional to its bandwidth. Assume that the rise time of a thermistor response is 0.05 s and the relation between the rise time and the bandwidth is $\tau_{\text{rise}} = 1/2\Delta f$. (A) Calculate the magnitude of the thermal noise. The ambient temperature is 27 °C and the thermistor resistance is 5 k Ω at this temperature. (B) What is the signal-to-noise ratio, if the average of current passing through the resistor is 0.2 mA?

Answer:

- (a) The bandwidth is equal to $\Delta f = 1/2\tau_{\text{rise}} = 1/2 \times 0.05 \text{ (s)} = 10 \text{ Hz}$ and according to (2.3), the rms value of the thermal noise voltage is equal to:

$$\begin{aligned}\bar{v}_{\text{rms}} &= \sqrt{4 \times 1.38 \times 10^{-23} \times 300 \text{ (K)} \times 5,000 \text{ (}\Omega\text{)} \times 10 \text{ (Hz)}} \\ &= 2.88 \times 10^{-8} \text{ (V)} = 0.0288 \text{ mV or } 20 \log(\bar{v}_{\text{rms}}) = -150.8 \text{ dB.}\end{aligned}$$

- (b) Current of 0.2 mA generates a voltage of $5,000 \text{ (k}\Omega\text{)} \times 0.0002 \text{ (A)} = 1 \text{ V}$ in the thermistor. As a result, the signal-to-noise ratio is:

$$\frac{S}{N} = 1\text{(V)}/2.88 \times 10^{-8}\text{(V)} = 3.47 \times 10^9.$$

2. *Shot noise*: The random fluctuations, which are caused by the carriers' random arrival time, produce shot noise. These signal carriers can be electrons, holes, photons, and phonons.

Shot noise is a random and quantized event, which depends on the transfer of the individual electrons across the junction. Using the statistical calculations, the root-mean-square of the current fluctuation, generated by the shot noise, can be obtained as:

$$\bar{i}_{\text{rms}} = \sqrt{2Ie\Delta f}, \quad (2.4)$$

where I is the average current passing through the junction, Δf is the bandwidth, and e is the charge of one electron, which is equal to 1.60×10^{-19} C.

Example 2. In a photodiode the bias current passing through the diode is 0.1 mA. (A) If the rise time of the photodiode is 0.2 ms and the relation between the rise time and the bandwidth is $\tau_{\text{rise}} = 1/4\Delta f$, calculate the rms value of the shot noise current fluctuation. (B) Calculate the magnitude of the shot noise voltage, when the junction resistance is equal to 250 Ω .

Answer: The bandwidth is equal to $\Delta f = 1/4\tau_{\text{rise}} = 1/[4 \times 0.0002 \text{ (s)}] = 1,250 \text{ Hz}$. According to (2.4) the rms value of the shot noise current fluctuation is equal to:

$$i_{\text{rms}} = \sqrt{2 \times 0.1 \times 10^{-3} \text{ (A)} \times 1.6 \times 10^{-19} \text{ (C)} \times 1,250 \text{ (Hz)}} = 200 \times 10^{-12} \text{ A} \\ = 200 \text{ pA.}$$

When the average resistance of the junction is equal to 250 Ω , this fluctuation current generates a rms voltage of $200 \times 10^{-12} \text{ (A)} \times 250 \text{ (}\Omega\text{)} = 50 \times 10^{-9} \text{ (V)}$ $= 0.05 \text{ }\mu\text{V}$ or $20 \log(\bar{v}_{\text{rms}}) = -146.02 \text{ dB}$.

3. *Generation-recombination* (or *g-r noise*): This type of noise is produced from the generation and recombination of electrons and holes in semiconductors. They are observed in junction electronic devices.
4. *Pink noise* (or *1/f noise*): In this type of noise the components of the frequency spectrum of the interfering signals are inversely proportional to the frequency. Pink noise is stronger at lower frequencies and each octave carries an equal amount of noise power. The origin of the pink signal is not completely understood.

رانش: اگر با وجود ثابت ماندن کمیت مورد اندازه گیری، خروجی به صورت پیوسته ایی تغییر کند، ویژگی رانش پاسخ حسگر مشاهده می شود. علت رانش ناپایداری حرارتی، ناپایداری مکانیکی، آلودگی ها و خرابی ماده حسگر است. معمولاً رانش را به **خط مبنا** مرتبط می کنند.

خط مبنا: مقدار خروجی حسگر بدون تحریک ورودی است. برای یک حسگر بدون رانش خط مبنا ثابت است.

2.2.8 Drift

Drift is observed when a gradual change in the sensing system's output is seen, while the measurand actually remains constant. Drift is the undesired change that is unrelated to the measurand. It is considered a systematic error, which can be attributed to interfering parameters such as mechanical instability and temperature instability, contamination, and the sensor's materials degradation. It is very common to assess the drift with respect to a sensor's *baseline*. Baseline is the output value, when the sensor is not exposed to a stimulus. Logically for a sensor with no drift, the baseline should remain constant.

2.2.9 Resolution

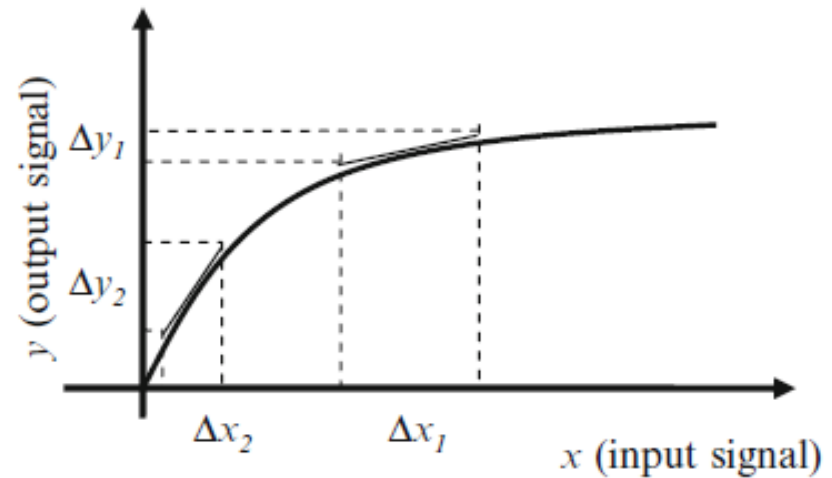
Resolution (or *discrimination*) is the minimal change of the measurand that can produce a detectable increment in the output signal. Resolution is strongly limited by any noise in the signal.

A temperature sensing system with four digits has a higher resolution than three digits. When the ambient temperature is 21 °C, the higher resolution system (four digits) output is 21.00 °C while the lower resolution system (three digits) is 21.0 °C. Obviously, the lower resolution system cannot resolve any values between 21.01 °C and 21.03 °C.

2.2.10 Minimum Detectable Signal

In a sensing system, *minimum detectable signal* (MDS) is the minimum signal increment that can be observed, when all interfering factors are taken into account. When the increment is assessed from zero, the value is generally referred to as *threshold* or *detection limit*. If the interferences are large relative to the input, it will be difficult to extract a clear signal and a small MDS cannot be obtained.

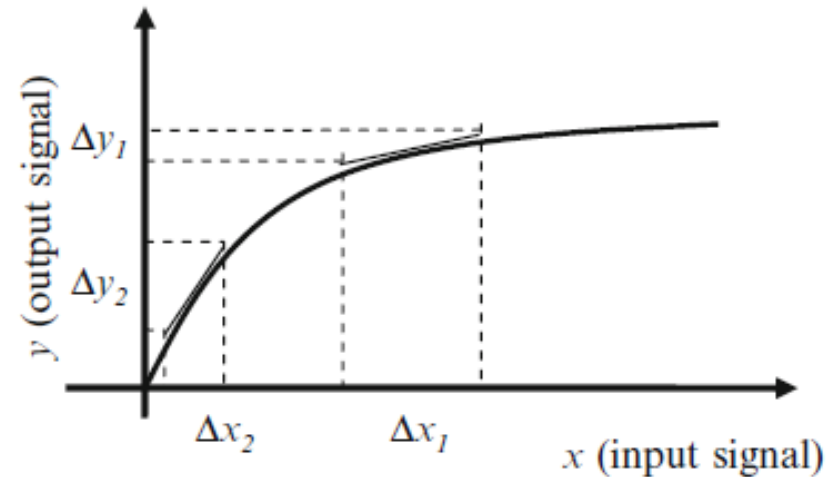
Fig. 2.2 Calibration curve:
it can be used for the
calculation of sensitivity



2.2.11 Calibration Curve

A sensing system has to be calibrated against a known measurand to assure that the sensing results in correct outputs. The relationship between the measured variable (x) and the signal variable generated by the system (y) is called a calibration curve as shown in Fig. 2.2.

Fig. 2.2 Calibration curve:
it can be used for the
calculation of sensitivity



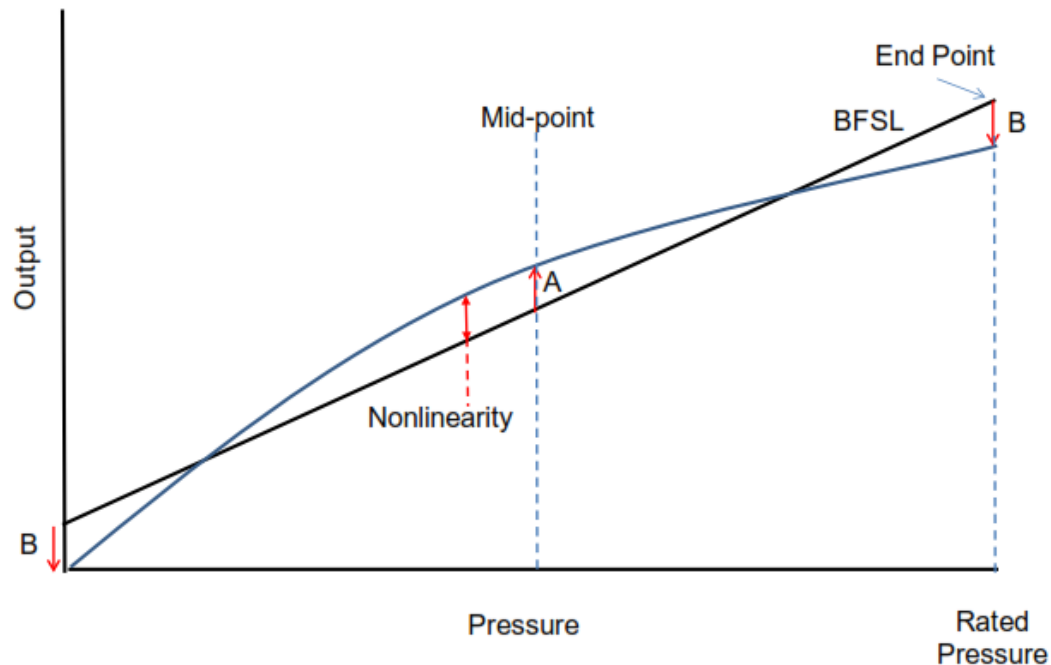
2.2.12 Sensitivity

Sensitivity is the ratio of the incremental change in the sensor's output (Δy) to the incremental change of the measurand in input (Δx). The slope of the calibration curve, $y = f(x)$, can be used for the calculation of sensitivity. As can be seen in Fig. 2.2, sensitivity can be altered depending on the calibration curve. In Fig. 2.2, the sensitivity for the lower values of the measurand ($\Delta y_1/\Delta x_1$) is larger than of the other section of the curve ($\Delta y_2/\Delta x_2$). An ideal sensor has a large and preferably constant sensitivity in its operating range. It is also seen that the sensor eventually reaches *saturation*, a state in which it can no longer respond to any changes.

2.2.13 Linearity

The closeness of the calibration curve to a specified straight line shows the *linearity* of a sensor. Its degree of resemblance to a straight line describes how linear a system is.

مثال: یک حسگر فشار با خروجی مقداری غیرخطی.



2.2.14 Selectivity

Selectivity is the sensing system's ability to measure a target measurand in the presence of others interferences.

For example, an oxygen gas sensor that does not show any response to other gas species, such as carbon dioxide or nitrogen oxide, is considered a very selective sensor.

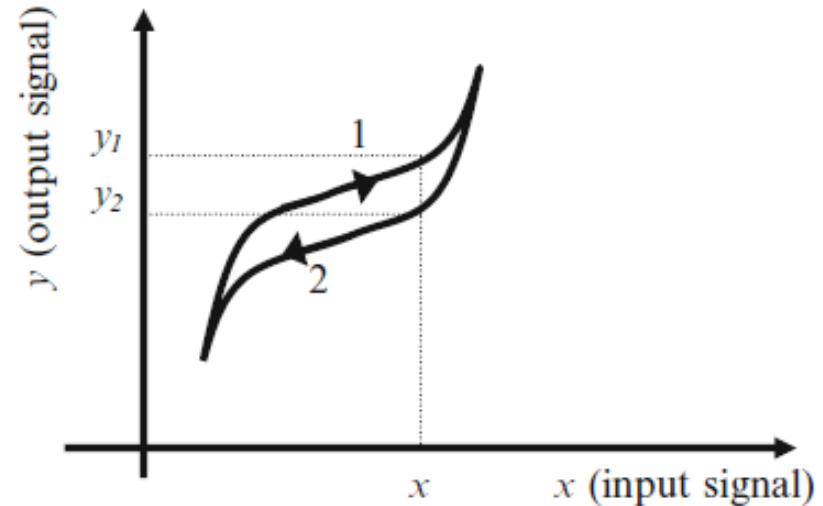
مثال: انتخابگری در یک حسگر گاز مقاومتی

Challenge gas	Concentration	R_G/R_A OR $-R_A/R_G$
NO ₂	10 ppm	200
NO	10 ppm	45
H ₂	100 ppm	-2.2
NH ₃	25 ppm	-1.1
CO ₂	5%	No response
SO ₂	5 ppm	No response
CH ₄	1%	No response

R_a: مقاومت حسگر مقاومتی در هوای تمیز.

R_g: مقاومت حسگر در هوای آلوده به گاز.

Fig. 2.3 An example of a hysteresis curve



2.2.15 Hysteresis

Hysteresis is the difference between output readings for the same measurand, depending on the trajectory followed by the sensor.

Hysteresis may cause false and inaccurate readings. Figure 2.3 represents the relation between output and input of a system with hysteresis. As can be seen, depending on whether path 1 or 2 is taken, two different outputs, for the same input, can be displayed by the sensing system.

2.2.16 Measurement Range

The maximum and minimum values of the measurand that can be measured with a sensing system are called the *measurement range*, which is also called *the dynamic range* or *span*. This range results in a meaningful and accurate output for the sensing system. All sensing systems are designed to perform over a specified range. Signals outside of this range may be unintelligible, cause unacceptably large inaccuracies, and may even result in irreversible damage to the sensor.

Generally the measurement range of a sensing system is specified on its technical sheet. For instance, if the measurement range of a temperature sensor is between -100 and 800°C , exposing it to temperatures outside this range may cause damage or generate inaccurate readings.

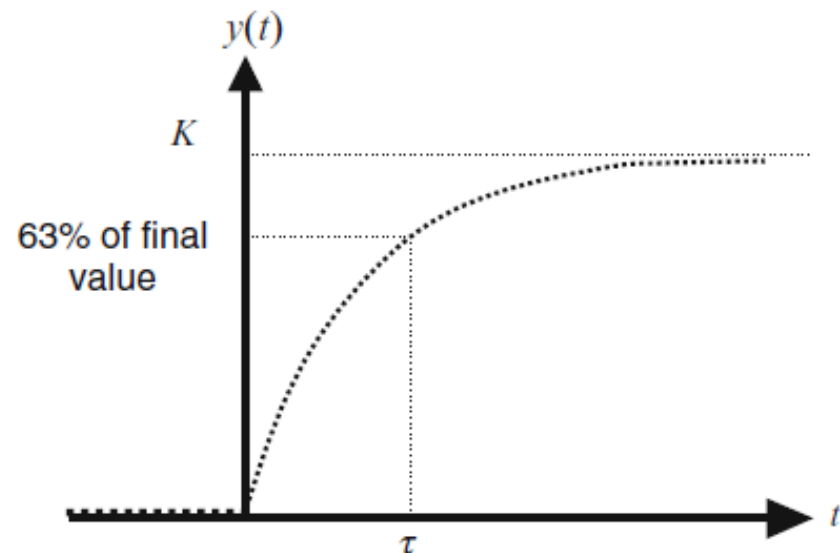
مثال: محدوده اندازه گیری برای سه نوع ترموکوپل:

Standard		Material	Temp Range
Calibration			
"J"	+	Iron	32 F to 1400 F
	-	Constantan	
"K"	+	Chromel	32 F to 2300 F
	-	Alumel	
"T"	+	Copper	-320 F to 700 F
	-	Constantan	

A sensing system response to a dynamically changing measurand can be quite different from when it is exposed to time invariable measurand. In the presence of a changing measurand, *dynamic characteristics* can be employed to describe the sensing system's transient properties. They can be used for defining how accurately the output signal is employed for the description of a time varying measurand. These characteristics deal with issues such as the rate at which the output changes in response to a measurand alteration and how these changes occur.

First-order system

Fig. 2.5 Response of a first-order system to a step function



Second-order system

$$G(s) = \frac{C(s)}{R(s)} = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$c_{step}(t) = K \left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin \left(\omega_d t + \tan^{-1} \left(\sqrt{1-\zeta^2} / \zeta \right) \right) \right]$$

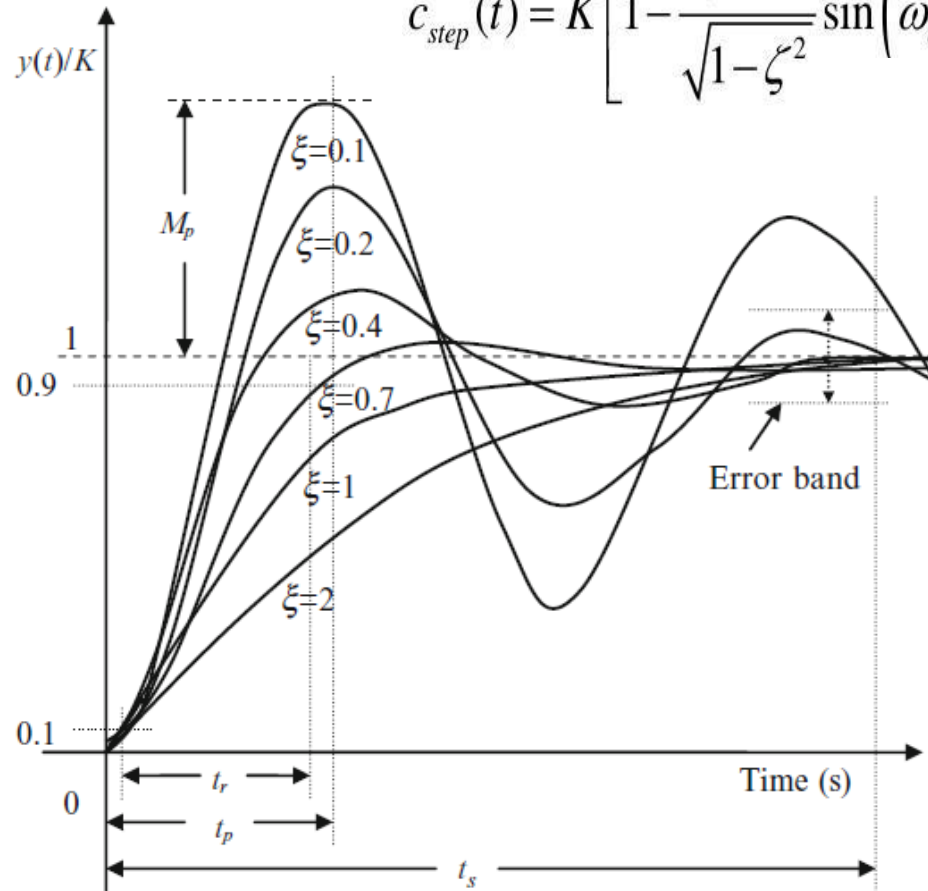
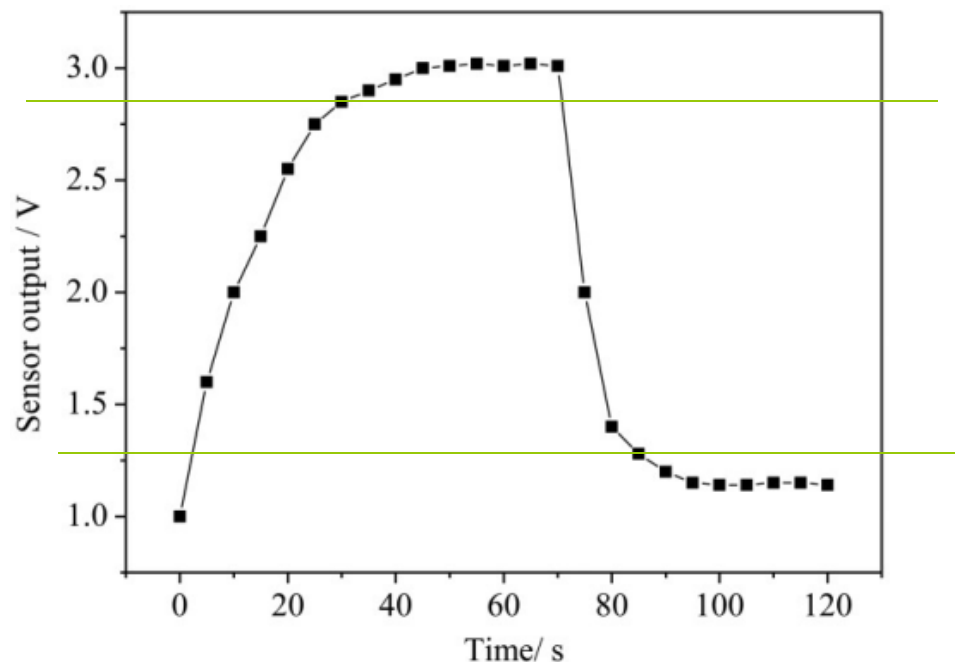


Fig. 2.8 Responses of a second-order sensing system to a step function at different damping ratios

2.2.17 Response and Recovery Time

When a sensing system is exposed to a measurand, the time it requires to reach a stable value is the response time. It is generally expressed as the time at which the output reaches a certain percentage (for instance, 95 %) of its final value, in response to a step change of the input. The “recovery time” is defined conversely.



**طبقه بندی حسگرها بر اساس پدیده های
فیزیکی که بر مبنای آنها عمل سنجش را
انجام می دهند.**

پدیده های فیزیکی که حسگرها بر اساس آنها کار می کنند:

- اثر فتوالکتریک
- اثر فتوکانداکتیو
- اثر فتو ولتاییک
- اثر دی الکتریک
- اثر فتو دی الکتریک
- اثر فتو لمینسانس
- اثر پرمبیلیتی (Permeability)
- اثر هال
- آثار ترموالکتریک (سیبک، پلتیر و تامسون)
- اثر ترمورزیستیو (Thermoresistive)

ادامه پدیده های فیزیکی که حسگرها بر اساس آنها کار می کنند:

- اثر پیزورزیستیو (Piezoresistive)
- اثر پیزوالکتریک (Piezoelectric)
- اثر پایروالکتریک (Pyroelectric)
- اثر مگنتورزیستنس (Magnetoresistance)
- اثر نرنست - اتینگشاوزن (Nernst/Ettingshausen)
- اثر داپلر (Doppler)
- اثر بارخهاوزن (Barkhausen)
- اثر فارادی
- اثر کر (Kerr)
- اثر پاکلز (Packels)

TI: Texas Instrument Company.

- حسگر دما
- حسگر موج میلی متری
- حسگر رطوبت
- حسگر نوری
- حسگر مغناطیسی
- حسگر دیود نوری
- حسگر خازنی
- حسگر القایی
- حسگر ماورا صوت
- حسگر گاز و شیمیایی
- مبدل LVDT
- حسگری جریان

تشکر از توجه شما

